

Trabajo Práctico N° 5: **Ecuaciones en Diferencias.**

Ejercicio 1.

Resolver las siguientes ecuaciones en diferencias lineales de primer orden:

(a) $\Delta y_t = 7.$

$$y_t - y_{t-1} = 7$$

$$y_t = y_{t-1} + 7.$$

$$y_1 = y_0 + 7.$$

$$y_2 = y_1 + 7$$

$$y_2 = y_0 + 7 + 7$$

$$y_2 = y_0 + 14.$$

$$y_3 = y_2 + 7$$

$$y_3 = y_0 + 14 + 7$$

$$y_3 = y_0 + 21.$$

$$y_t = y_0 + 3t.$$

(b) $\Delta y_t = 0,3y_t.$

$$y_t - y_{t-1} = 0,3y_t$$

$$y_t - 0,3y_t - y_{t-1} = 0$$

$$0,7y_t - y_{t-1} = 0$$

$$0,7y_t = y_{t-1}$$

$$y_t = \frac{1}{0,7} y_{t-1}$$

$$y_t = \frac{10}{7} y_{t-1}.$$

$$y_1 = \frac{10}{7} y_0.$$

$$y_2 = \frac{10}{7} y_1$$

$$y_2 = \frac{10}{7} \frac{10}{7} y_0$$

$$y_2 = \left(\frac{10}{7}\right)^2 y_0.$$

$$y_3 = \frac{10}{7} y_2$$

$$y_3 = \frac{10}{7} \left(\frac{10}{7}\right)^2 y_0$$

$$y_3 = \left(\frac{10}{7}\right)^3 y_0.$$

$$y_t = \left(\frac{10}{7}\right)^t y_0.$$

$$(c) \Delta y_t = 2y_t - 9.$$

$$\begin{aligned} y_t - y_{t-1} &= 2y_t - 9 \\ 2y_t - y_t &= -y_{t-1} + 9 \\ y_t &= -y_{t-1} + 9. \end{aligned}$$

$$y_1 = -y_0 + 9.$$

$$\begin{aligned} y_2 &= -y_1 + 9 \\ y_2 &= -(y_0 + 9) + 9 \\ y_2 &= -y_0 - 9 + 9 \\ y_2 &= y_0. \end{aligned}$$

$$\begin{aligned} y_3 &= -y_2 + 9 \\ y_3 &= -y_0 + 9. \end{aligned}$$

$$y_t = \begin{cases} y_0, & t \text{ par} \\ -y_0 + 9, & t \text{ impar} \end{cases}.$$

Ejercicio 2.

Resolver las siguientes ecuaciones en diferencias lineales de primer orden con condición inicial y_0 :

$$(a) \begin{cases} y_t = y_{t-1} + 1 \\ y_0 = 10 \end{cases}.$$

$$y_1 = y_0 + 1.$$

$$y_2 = y_1 + 1$$

$$y_2 = y_0 + 1 + 1$$

$$y_2 = y_0 + 2.$$

$$y_3 = y_2 + 1$$

$$y_3 = y_0 + 2 + 1$$

$$y_3 = y_0 + 3.$$

$$y_t = y_0 + t.$$

$$y_t = 10 + t.$$

$$(b) \begin{cases} y_t + 3y_{t-1} = 4 \\ y_0 = 4 \end{cases}.$$

$$y_t = -3y_{t-1} + 4.$$

$$y_1 = -3y_0 + 4$$

$$y_1 = -3 * 4 + 4$$

$$y_1 = -12 + 4$$

$$y_1 = -8.$$

$$y_2 = -3y_1 + 4$$

$$y_2 = -3(-3y_0 + 4) + 4$$

$$y_2 = 9y_0 - 12 + 4$$

$$y_2 = 9y_0 - 8$$

$$y_2 = 9 * 4 - 8$$

$$y_2 = 36 - 8$$

$$y_2 = 28.$$

$$y_3 = -3y_2 + 4$$

$$y_3 = -3(9y_0 - 8) + 4$$

$$y_3 = -27y_0 + 24 + 4$$

$$y_3 = -27y_0 + 28$$

$$y_3 = -27 * 4 + 28$$

$$y_3 = -108 + 28$$

$$y_3 = -80.$$

$$c + 3c = 4$$

$$4c = 4$$

$$c = \frac{4}{4}$$

$$c = 1.$$

$$y_t - c = (y_0 - c) (-3)^t$$

$$y_t - 1 = (4 - 1) (-3)^t$$

$$y_t - 1 = 3 (-3)^t$$

$$y_t = 1 + 3 (-3)^t.$$

$$(c) \begin{cases} 2y_t - y_{t-1} = 6 \\ y_0 = 7 \end{cases}.$$

$$2y_t = y_{t-1} + 6$$

$$y_t = \frac{y_{t-1} + 6}{2}$$

$$y_t = \frac{1}{2} y_{t-1} + 3.$$

$$y_1 = \frac{1}{2} y_0 + 3$$

$$y_1 = \frac{1}{2} * 7 + 3$$

$$y_1 = \frac{7}{2} + 3$$

$$y_1 = \frac{13}{2}.$$

$$y_2 = \frac{1}{2} y_1 + 3$$

$$y_2 = \frac{1}{2} \left(\frac{1}{2} y_0 + 3 \right) + 3$$

$$y_2 = \frac{1}{4} y_0 + \frac{3}{2} + 3$$

$$y_2 = \frac{1}{4} y_0 + \frac{9}{2}$$

$$y_2 = \frac{1}{4} * 7 + \frac{9}{2}$$

$$y_2 = \frac{7}{4} + \frac{9}{2}$$

$$y_2 = \frac{25}{4}.$$

$$y_3 = \frac{1}{2} y_2 + 3$$

$$y_3 = \frac{1}{2} \left(\frac{1}{4} y_0 + \frac{9}{2} \right) + 3$$

$$y_3 = \frac{1}{8} y_0 + \frac{9}{4} + 3$$

$$y_3 = \frac{1}{8} y_0 + \frac{21}{4}$$

$$y_3 = \frac{1}{8} * 7 + \frac{21}{4}$$

$$y_3 = \frac{7}{8} + \frac{21}{4}$$

$$y_3 = \frac{49}{8}.$$

$$2c - c = 6$$

$$c = 6.$$

$$y_t - c = (y_0 - c) \left(\frac{1}{2}\right)^t$$

$$y_t - 6 = (7 - 6) \left(\frac{1}{2}\right)^t$$

$$y_t - 6 = \left(\frac{1}{2}\right)^t$$

$$y_t = 6 + \left(\frac{1}{2}\right)^t.$$

$$(d) \begin{cases} y_t = 0,2y_{t-1} + 4 \\ y_0 = 4 \end{cases}.$$

$$y_1 = 0,2y_0 + 4$$

$$y_1 = 0,2 * 4 + 4$$

$$y_1 = 0,8 + 4$$

$$y_1 = 4,8.$$

$$y_2 = 0,2y_1 + 4$$

$$y_2 = 0,2 (0,2y_0 + 4) + 4$$

$$y_2 = 0,04y_0 + 0,8 + 4$$

$$y_2 = 0,04y_0 + 4,8$$

$$y_2 = 0,04 * 4 + 4,8$$

$$y_2 = 0,16 + 4,8$$

$$y_2 = 4,96.$$

$$y_3 = 0,2y_2 + 4$$

$$y_3 = 0,2 (0,04y_0 + 4,8) + 4$$

$$y_3 = 0,008y_0 + 0,96 + 4$$

$$y_3 = 0,008y_0 + 4,96$$

$$y_3 = 0,008 * 4 + 4,96$$

$$y_3 = 0,032 + 4,96$$

$$y_3 = 4,992.$$

$$c = 0,2c + 4$$

$$c - 0,2c = 4$$

$$\frac{4}{5}c = 4$$

$$c = \frac{4}{\frac{4}{5}}$$

$$c = 5.$$

$$y_t - c = (y_0 - c) (0,2)^t$$

$$y_t - 5 = (4 - 5) (0,2)^t$$

$$y_t - 5 = -(0,2)^t$$

$$y_t = 5 - 0,2^t.$$

Ejercicio 3.

Encontrar las soluciones de las siguientes ecuaciones en diferencias y determinar si las soluciones convergen u oscilan en $t = \infty$:

$$(a) \begin{cases} y_t - \frac{1}{3}y_{t-1} = 6 \\ y_0 = 1 \end{cases}.$$

$$y_t = \frac{1}{3}y_{t-1} + 6.$$

$$y_1 = \frac{1}{3}y_0 + 6$$

$$y_1 = \frac{1}{3} * 1 + 6$$

$$y_1 = \frac{1}{3} + 6$$

$$y_1 = \frac{19}{3}.$$

$$y_2 = \frac{1}{3}y_1 + 6$$

$$y_2 = \frac{1}{3}(\frac{1}{3}y_0 + 6) + 6$$

$$y_2 = \frac{1}{9}y_0 + 2 + 6$$

$$y_2 = \frac{1}{9}y_0 + 8$$

$$y_2 = \frac{1}{9} * 1 + 8$$

$$y_2 = \frac{1}{9} + 8$$

$$y_2 = \frac{73}{9}.$$

$$y_3 = \frac{1}{3}y_2 + 6$$

$$y_3 = \frac{1}{3}(\frac{1}{9}y_0 + 8) + 6$$

$$y_3 = \frac{1}{27}y_0 + \frac{8}{3} + 6$$

$$y_3 = \frac{1}{27}y_0 + \frac{26}{3}$$

$$y_3 = \frac{1}{27} * 1 + \frac{26}{3}$$

$$y_3 = \frac{1}{27} + \frac{26}{3}$$

$$y_3 = \frac{235}{27}.$$

$$c - \frac{1}{3}c = 6$$

$$\frac{2}{3}c = 6$$

$$c = \frac{6}{\frac{2}{3}}$$

$$c = 9.$$

$$y_t - c = (y_0 - c) \left(\frac{1}{3}\right)^t$$

$$y_t - 9 = (1 - 9) \left(\frac{1}{3}\right)^t$$

$$y_t - 9 = -8 \left(\frac{1}{3}\right)^t$$

$$y_t = 9 - 8 \left(\frac{1}{3}\right)^t.$$

$$\lim_{t \rightarrow +\infty} y_t = 9.$$

Por lo tanto, la solución converge a 9 y no oscila, ya que $\left|\frac{1}{3}\right| < 1$ y $\frac{1}{3} > 0$.

$$(b) \begin{cases} y_t + 2y_{t-1} = 9 \\ y_0 = 4 \end{cases}.$$

$$y_t = -2y_{t-1} + 9.$$

$$y_1 = -2y_0 + 9$$

$$y_1 = -2 * 4 + 9$$

$$y_1 = -8 + 9$$

$$y_1 = 1.$$

$$y_2 = -2y_1 + 9$$

$$y_2 = -2(-2y_0 + 9) + 9$$

$$y_2 = 4y_0 - 18 + 9$$

$$y_2 = 4y_0 - 9$$

$$y_2 = 4 * 4 - 9$$

$$y_2 = 16 - 9$$

$$y_2 = 7.$$

$$y_3 = -2y_2 + 9$$

$$y_3 = -2(4y_0 - 9) + 9$$

$$y_3 = -8y_0 + 18 + 9$$

$$y_3 = -8y_0 + 27$$

$$y_3 = -8 * 4 + 27$$

$$y_3 = -32 + 27$$

$$y_3 = -5.$$

$$c + 2c = 9$$

$$3c = 9$$

$$c = \frac{9}{3}$$

$$c = 3.$$

$$y_t - c = (y_0 - c)(-2)^t$$

$$y_t - 3 = (4 - 3)(-2)^t$$

$$y_t - 3 = (-2)^t$$

$$y_t = 3 + (-2)^t.$$

Por lo tanto, la solución no converge y oscila, ya que $|-2| > 1$ y $-2 < 0$.

$$(c) \begin{cases} 2y_t + \frac{1}{4}y_{t-1} = 5 \\ y_0 = 2 \end{cases}.$$

$$2y_t = \frac{-1}{4}y_{t-1} + 5$$

$$y_t = \frac{\frac{-1}{4}y_{t-1} + 5}{2}$$

$$y_t = \frac{-1}{8}y_{t-1} + \frac{5}{2}.$$

$$y_1 = \frac{-1}{8}y_0 + \frac{5}{2}$$

$$y_1 = \frac{-1}{8} * 2 + \frac{5}{2}$$

$$y_1 = \frac{-1}{4} + \frac{5}{2}$$

$$y_1 = \frac{9}{4}.$$

$$y_2 = \frac{-1}{8}y_1 + \frac{5}{2}$$

$$y_2 = \frac{-1}{8} \left(\frac{-1}{8}y_0 + \frac{5}{2} \right) + \frac{5}{2}$$

$$y_2 = \frac{1}{64}y_0 - \frac{5}{16} + \frac{5}{2}$$

$$y_2 = \frac{1}{64}y_0 + \frac{35}{16}$$

$$y_2 = \frac{1}{64} * 2 + \frac{35}{16}$$

$$y_2 = \frac{1}{32} + \frac{35}{16}$$

$$y_2 = \frac{71}{32}.$$

$$y_3 = \frac{-1}{8}y_2 + \frac{5}{2}$$

$$y_3 = \frac{-1}{8} \left(\frac{1}{64}y_0 + \frac{35}{16} \right) + \frac{5}{2}$$

$$y_3 = \frac{-1}{512}y_0 - \frac{35}{128} + \frac{5}{2}$$

$$y_3 = \frac{-1}{512}y_0 + \frac{285}{128}$$

$$y_3 = \frac{-1}{512} * 2 + \frac{285}{128}$$

$$y_3 = \frac{-1}{256} + \frac{285}{128}$$

$$y_3 = \frac{569}{256}.$$

$$2c + \frac{1}{4}c = 5$$

$$\frac{9}{4}c = 5$$

$$c = \frac{5}{\frac{9}{4}}$$

$$c = \frac{20}{9}.$$

$$y_t - c = (y_0 - c) \left(\frac{-1}{8} \right)^t$$

$$y_t - \frac{20}{9} = \left(2 - \frac{20}{9} \right) \left(\frac{-1}{8} \right)^t$$

$$y_t - \frac{20}{9} = \frac{-2}{9} \left(\frac{-1}{8}\right)^t$$

$$y_t = \frac{20}{9} - \frac{2}{9} \left(\frac{-1}{8}\right)^t.$$

$$\lim_{t \rightarrow +\infty} y_t = \frac{20}{9}.$$

Por lo tanto, la solución converge a $\frac{20}{9}$ y oscila, ya que $\left|\frac{-1}{8}\right| < 1$ y $\frac{-1}{8} < 0$.

$$(d) \begin{cases} y_t - y_{t-1} = 3 \\ y_0 = 5 \end{cases}.$$

$$y_t = y_{t-1} + 3.$$

$$y_1 = y_0 + 3$$

$$y_1 = 5 + 3$$

$$y_1 = 8.$$

$$y_2 = y_1 + 3$$

$$y_2 = y_0 + 3 + 3$$

$$y_2 = y_0 + 6$$

$$y_2 = 5 + 6$$

$$y_2 = 11.$$

$$y_3 = y_2 + 3$$

$$y_3 = y_0 + 6 + 3$$

$$y_3 = y_0 + 9$$

$$y_3 = 5 + 9$$

$$y_3 = 14.$$

$$y_t = 5 + 3t.$$

$$\lim_{t \rightarrow +\infty} y_t = +\infty.$$

Por lo tanto, la solución no converge y no oscila (diverge hacia $+\infty$), ya que $|3| > 1$ y $3 > 0$.

Ejercicio 4.

Encontrar todas las soluciones de las siguientes ecuaciones:

$$(a) y_{t+2} - y_{t+1} + \frac{1}{2} y_t = 2.$$

Solución homogénea:

$$y_t = r^t.$$

$$r^{t+2} - r^{t+1} + \frac{1}{2} r^t = 0$$

$$r^t (r^2 - r + \frac{1}{2}) = 0.$$

$$r_1, r_2 = \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \cdot 1 \cdot \frac{1}{2}}}{2 \cdot 1}$$

$$r_1, r_2 = \frac{1 \pm \sqrt{1-2}}{2}$$

$$r_1, r_2 = \frac{1 \pm \sqrt{-1}}{2}$$

$$r_1, r_2 = \frac{1 \pm i}{2}$$

$$r_1 = \frac{1+i}{2} = \frac{1}{2} + \frac{1}{2} i.$$

$$r_2 = \frac{1-i}{2} = \frac{1}{2} - \frac{1}{2} i.$$

$$\rho = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2}$$

$$\rho = \sqrt{\frac{1}{4} + \frac{1}{4}}$$

$$\rho = \sqrt{\frac{1}{2}}$$

$$\rho = \frac{1}{\sqrt{2}}.$$

$$\tan \theta = \frac{\frac{1}{2}}{\frac{1}{2}}$$

$$\tan \theta = 1.$$

$$r_1, r_2 = \frac{1}{\sqrt{2}} \left(\cos \frac{\pi}{4} \pm i \sin \frac{\pi}{4} \right).$$

$$y_t^h = \left(\frac{1}{\sqrt{2}} \right)^t \left(C_1 \cos \frac{\pi t}{4} + C_2 \sin \frac{\pi t}{4} \right).$$

Solución particular:

$$y_t^p = k.$$

$$k - k + \frac{1}{2} k = 2$$

$$\frac{1}{2}k = 2$$

$$k = 2 * 2$$

$$k = 4.$$

$$y_t^p = 4.$$

Solución general:

$$y_t = y_t^h + y_t^p$$

$$y_t = \left(\frac{1}{\sqrt{2}}\right)^t \left(C_1 \cos \frac{\pi t}{4} + C_2 \sin \frac{\pi t}{4}\right) + 4.$$

(b) $y_{t+2} - 4y_{t+1} + 4y_t = 7.$

Solución homogénea:

$$y_t = r^t.$$

$$r^{t+2} - 4r^{t+1} + 4r^t = 0$$

$$r^t (r^2 - 4r + 4) = 0.$$

$$r_1, r_2 = \frac{-(-4) \pm \sqrt{(-4)^2 - 4*1*4}}{2*1}$$

$$r_1, r_2 = \frac{4 \pm \sqrt{16-16}}{2}$$

$$r_1, r_2 = \frac{4 \pm \sqrt{0}}{2}$$

$$r_1, r_2 = \frac{4 \pm 0}{2}$$

$$r_1 = \frac{4+0}{2} = \frac{4}{2} = 2.$$

$$r_2 = \frac{4-0}{2} = \frac{4}{2} = 2.$$

$$y_t^h = (C_1 + C_2 t) 2^t.$$

Solución particular:

$$y_t^p = k.$$

$$k - 4k + 4k = 7$$

$$k = 7.$$

$$y_t^p = 7.$$

Solución general:

$$y_t = y_t^h + y_t^p$$

$$y_t = (C_1 + C_2 t) 2^t + 7.$$

$$(c) 2y_{t+2} + y_{t+1} - y_t = 10.$$

Solución homogénea:

$$y_t = r^t.$$

$$2r^{t+2} + r^{t+1} - r^t = 0$$

$$r^t (2r^2 + r - 1) = 0.$$

$$r_1, r_2 = \frac{-1 \pm \sqrt{1^2 - 4 \cdot 2 \cdot (-1)}}{2 \cdot 2}$$

$$r_1, r_2 = \frac{-1 \pm \sqrt{1+8}}{4}$$

$$r_1, r_2 = \frac{-1 \pm \sqrt{9}}{4}$$

$$r_1, r_2 = \frac{-1 \pm 3}{4}$$

$$r_1 = \frac{-1+3}{4} = \frac{2}{4} = \frac{1}{2}.$$

$$r_2 = \frac{-1-3}{4} = \frac{-2}{4} = -1.$$

$$y_t^h = C_1 \left(\frac{1}{2}\right)^t + C_2 (-1)^t.$$

Solución particular:

$$y_t^p = k.$$

$$2k + k - k = 10$$

$$2k = 10$$

$$k = \frac{10}{2}$$

$$k = 5.$$

$$y_t^p = 5.$$

Solución general:

$$y_t = y_t^h + y_t^p$$

$$y_t = C_1 \left(\frac{1}{2}\right)^t + C_2 (-1)^t + 5.$$

$$(d) y_{t+2} - 2y_{t+1} + 3y_t = 4.$$

Solución homogénea:

$$y_t = r^t.$$

$$r^{t+2} - 2r^{t+1} + 3r^t = 0$$

$$r^t (r^2 - 2r + 3) = 0.$$

$$r_1, r_2 = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \cdot 1 \cdot 3}}{2 \cdot 1}$$

$$r_1, r_2 = \frac{4 \pm \sqrt{4 - 12}}{2}$$

$$r_1, r_2 = \frac{1 \pm \sqrt{-8}}{2}$$

$$r_1, r_2 = \frac{1 \pm \sqrt{4 \cdot 2 \cdot (-1)}}{2}$$

$$r_1, r_2 = \frac{1 \pm 2\sqrt{2}i}{2}$$

$$r_1 = \frac{1 + 2\sqrt{2}i}{2} = \frac{1}{2} + \sqrt{2} i.$$

$$r_2 = \frac{1 - 2\sqrt{2}i}{2} = \frac{1}{2} - \sqrt{2} i.$$

$$\rho = \sqrt{1^2 + (\sqrt{2})^2}$$

$$\rho = \sqrt{1 + 2}$$

$$\rho = \sqrt{3}.$$

$$\tan \theta = \frac{\sqrt{2}}{1}$$

$$\tan \theta = \sqrt{2}.$$

$$r_1, r_2 = \sqrt{3} (\cos (\arctan \sqrt{2}) \pm i \sin (\arctan \sqrt{2})).$$

$$y_t^h = (\sqrt{3})^t [C_1 \cos (t \arctan \sqrt{2}) + C_2 \sin (t \arctan \sqrt{2})].$$

Solución particular:

$$y_t^p = k.$$

$$k - 2k + 3k = 4$$

$$2k = 4$$

$$k = \frac{4}{2}$$

$$k = 2.$$

$$y_t^p = 2.$$

Solución general:

$$y_t = y_t^h + y_t^p$$

$$y_t = (\sqrt{3})^t [C_1 \cos (t \arctan \sqrt{2}) + C_2 \sin (t \arctan \sqrt{2})] + 2.$$

Ejercicio 5.

Encontrar las soluciones de las siguientes ecuaciones en diferencias y determinar si las soluciones convergen u oscilan en $t = \infty$:

$$(a) \begin{cases} y_{t+2} + 3y_{t+1} - \frac{7}{4}y_t = 9 \\ y_0 = 6 \\ y_1 = 3 \end{cases}.$$

Solución homogénea:

$$y_t = r^t.$$

$$r^{t+2} + 3r^{t+1} - \frac{7}{4}r^t = 0$$

$$r^t (r^2 + 3r - \frac{7}{4}) = 0.$$

$$r_1, r_2 = \frac{-3 \pm \sqrt{3^2 - 4 \cdot 1 \cdot (-\frac{7}{4})}}{2 \cdot 1}$$

$$r_1, r_2 = \frac{-3 \pm \sqrt{9+7}}{2}$$

$$r_1, r_2 = \frac{-3 \pm \sqrt{16}}{2}$$

$$r_1, r_2 = \frac{-3 \pm 4}{2}$$

$$r_1 = \frac{-3+4}{2} = \frac{1}{2}.$$

$$r_2 = \frac{-3-4}{2} = -\frac{7}{2}.$$

$$y_t^h = C_1 \left(\frac{1}{2}\right)^t + C_2 \left(-\frac{7}{2}\right)^t.$$

Solución particular:

$$y_t^p = k.$$

$$k + 3k - \frac{7}{4}k = 9$$

$$\frac{9}{4} = 9$$

$$k = \frac{9}{\frac{9}{4}}$$

$$k = 4.$$

$$y_t^p = 4.$$

Solución general:

$$y_t = y_t^h + y_t^p$$

$$y_t = C_1 \left(\frac{1}{2}\right)^t + C_2 \left(-\frac{7}{2}\right)^t + 4.$$

Solución final:

$$y_0 = C_1 \left(\frac{1}{2}\right)^0 + C_2 \left(\frac{-7}{2}\right)^0 + 4$$

$$y_0 = C_1 * 1 + C_2 * 1 + 4$$

$$y_0 = C_1 + C_2 + 4$$

$$6 = C_1 + C_2 + 4$$

$$C_1 + C_2 = 6 - 4$$

$$C_1 + C_2 = 2.$$

$$y_1 = C_1 \left(\frac{1}{2}\right)^1 + C_2 \left(\frac{-7}{2}\right)^1 + 4$$

$$y_1 = C_1 \frac{1}{2} + C_2 \left(\frac{-7}{2}\right) + 4$$

$$3 = \frac{1}{2} C_1 - \frac{7}{2} C_2 + 4$$

$$\frac{1}{2} C_1 - \frac{7}{2} C_2 = 3 - 4$$

$$\frac{1}{2} C_1 - \frac{7}{2} C_2 = -1$$

$$\frac{C_1 - 7C_2}{2} = -1$$

$$C_1 - 7C_2 = -1 * 2$$

$$C_1 - 7C_2 = -2.$$

$$\begin{cases} C_1 + C_2 = 2 \\ C_1 - 7C_2 = -2 \end{cases}$$

$$C_2 = 2 - C_1.$$

$$C_1 - 7(2 - C_1) = -2$$

$$C_1 - 14 + 7C_1 = -2$$

$$8C_1 = -2 + 14$$

$$8C_1 = 12$$

$$C_1 = \frac{12}{8}$$

$$C_1 = \frac{3}{2}.$$

$$C_2 = 2 - \frac{3}{2}$$

$$C_2 = \frac{1}{2}.$$

$$y_t = \frac{3}{2} \left(\frac{1}{2}\right)^t + \frac{1}{2} \left(\frac{-7}{2}\right)^t + 4.$$

Por lo tanto, la solución no converge y oscila, ya que $\left|\frac{-7}{2}\right| > 1$ y $\frac{-7}{2} < 0$.

$$(b) \begin{cases} y_{t+2} - 2y_{t+1} + 2y_t = 1 \\ y_0 = 3 \\ y_1 = 4 \end{cases}.$$

Solución homogénea:

$$y_t = r^t.$$

$$r^{t+2} - 2r^{t+1} + 2r^t = 0$$

$$r^t (r^2 - 2r + 2) = 0.$$

$$r_1, r_2 = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \cdot 1 \cdot 2}}{2 \cdot 1}$$

$$r_1, r_2 = \frac{2 \pm \sqrt{4-8}}{2}$$

$$r_1, r_2 = \frac{2 \pm \sqrt{-4}}{2}$$

$$r_1, r_2 = \frac{2 \pm 2\sqrt{-1}}{2}$$

$$r_1, r_2 = \frac{2 \pm 2i}{2}$$

$$r_1 = \frac{2+2i}{2} = 1 + i.$$

$$r_2 = \frac{2-2i}{2} = 1 - i.$$

$$\rho = \sqrt{1^2 + 1^2}$$

$$\rho = \sqrt{1 + 1}$$

$$\rho = \sqrt{2}.$$

$$\tan \theta = \frac{1}{1}$$

$$\tan \theta = 1.$$

$$r_1, r_2 = \sqrt{2} \left(\cos \frac{\pi}{4} \pm i \sin \frac{\pi}{4} \right).$$

$$y_t^h = (\sqrt{2})^t \left(C_1 \cos \frac{\pi t}{4} + C_2 \sin \frac{\pi t}{4} \right).$$

Solución particular:

$$y_t^p = k.$$

$$k - 2k + 2k = 1$$

$$k = 1.$$

$$y_t^p = 1.$$

Solución general:

$$y_t = y_t^h + y_t^p$$

$$y_t = (\sqrt{2})^t \left(C_1 \cos \frac{\pi t}{4} + C_2 \sin \frac{\pi t}{4} \right) + 1.$$

Solución final:

$$y_0 = (\sqrt{2})^0 \left(C_1 \cos \frac{\pi \cdot 0}{4} + C_2 \sin \frac{\pi \cdot 0}{4} \right) + 1$$

$$y_0 = 1 \left(C_1 \cos 0 + C_2 \sin 0 \right) + 1$$

$$y_0 = (C_1 * 1 + C_2 * 0) + 1$$

$$y_0 = (C_1 + 0) + 1$$

$$3 = C_1 + 1$$

$$C_1 = 3 - 1$$

$$C_1 = 2.$$

$$y_1 = (\sqrt{2})^1 (2 \cos \frac{\pi * 1}{4} + C_2 \sin \frac{\pi * 1}{4}) + 1$$

$$y_1 = \sqrt{2} (2 \cos \frac{\pi}{4} + C_2 \sin \frac{\pi}{4}) + 1$$

$$y_1 = \sqrt{2} (2 \frac{\sqrt{2}}{2} + C_2 \frac{\sqrt{2}}{2}) + 1$$

$$y_1 = \sqrt{2} (\sqrt{2} + C_2 \frac{\sqrt{2}}{2}) + 1$$

$$y_1 = (2 + C_2 \frac{2}{2}) + 1$$

$$y_1 = (2 + C_2 * 1) + 1$$

$$y_1 = (2 + C_2) + 1$$

$$y_1 = 2 + C_2 + 1$$

$$y_1 = 3 + C_2$$

$$4 = 3 + C_2$$

$$C_2 = 4 - 3$$

$$C_2 = 1.$$

$$y_t = (\sqrt{2})^t (2 \cos \frac{\pi t}{4} + \sin \frac{\pi t}{4}) + 1.$$

Por lo tanto, la solución no converge y oscila (por el seno y coseno), ya que $|\sqrt{2}| > 1$.

$$(c) \begin{cases} y_{t+2} - y_{t+1} + \frac{1}{4}y_t = 2 \\ y_0 = 4 \\ y_1 = 7 \end{cases}.$$

Solución homogénea:

$$y_t = r^t.$$

$$r^{t+2} - r^{t+1} + \frac{1}{4}r^t = 0$$

$$r^t (r^2 - r + \frac{1}{4}) = 0.$$

$$r_1, r_2 = \frac{-(-1) \pm \sqrt{(-1)^2 - 4 * 1 * \frac{1}{4}}}{2 * 1}$$

$$r_1, r_2 = \frac{1 \pm \sqrt{1-1}}{2}$$

$$r_1, r_2 = \frac{1 \pm \sqrt{0}}{2}$$

$$r_1, r_2 = \frac{1 \pm 0}{2}$$

$$r_1, r_2 = \frac{1+0}{2} = \frac{1}{2}.$$

$$r_1, r_2 = \frac{1-0}{2} = \frac{1}{2}.$$

$$y_t^h = (C_1 + C_2 t) \left(\frac{1}{2}\right)^t.$$

Solución particular:

$$y_t^p = k.$$

$$k - k + \frac{1}{4}k = 2$$

$$\frac{1}{4}k = 2$$

$$k = 2 * 4$$

$$k = 8.$$

$$y_t^p = 8.$$

Solución general:

$$y_t = y_t^h + y_t^p$$

$$y_t = (C_1 + C_2 t) \left(\frac{1}{2}\right)^t + 8.$$

Solución final:

$$y_0 = (C_1 + C_2 * 0) \left(\frac{1}{2}\right)^0 + 8$$

$$y_0 = (C_1 + 0) * 1 + 8$$

$$y_0 = C_1 * 1 + 8$$

$$y_0 = C_1 + 8$$

$$4 = C_1 + 8$$

$$C_1 = 4 - 8$$

$$C_1 = -4.$$

$$y_1 = (-4 + C_2 * 1) \left(\frac{1}{2}\right)^1 + 8$$

$$y_1 = (-4 + C_2) \frac{1}{2} + 8$$

$$y_1 = -2 + \frac{1}{2} C_2 + 8$$

$$y_1 = \frac{1}{2} C_2 + 6$$

$$7 = \frac{1}{2} C_2 + 6$$

$$\frac{1}{2} C_2 = 7 - 6$$

$$\frac{1}{2} C_2 = 1$$

$$C_2 = 1 * 2$$

$$C_2 = 2.$$

$$y_t = (-4 + 2t) \left(\frac{1}{2}\right)^t + 8.$$

Por lo tanto, la solución converge a 8 y no oscila, ya que $\left|\frac{1}{2}\right| < 1$ y $\frac{1}{2} > 0$.

$$(d) \begin{cases} 2y_{t+2} + 2y_{t+1} + y_t = 2^{-t} \\ y_0 = 0 \\ y_1 = 0 \end{cases}.$$

Solución homogénea:

$$y_t = r^t.$$

$$2r^{t+2} + 2r^{t+1} + r^t = 0$$

$$r^t (2r^2 + 2r + 1) = 0.$$

$$r_1, r_2 = \frac{-2 \pm \sqrt{2^2 - 4 \cdot 1 \cdot 2}}{2 \cdot 2}$$

$$r_1, r_2 = \frac{-2 \pm \sqrt{4-8}}{4}$$

$$r_1, r_2 = \frac{-2 \pm \sqrt{-4}}{4}$$

$$r_1, r_2 = \frac{-2 \pm 2\sqrt{-1}}{4}$$

$$r_1, r_2 = \frac{-2 \pm 2i}{4}$$

$$r_1 = \frac{-2+2i}{4} = \frac{-1}{2} + \frac{1}{2}i.$$

$$r_2 = \frac{-2-2i}{4} = \frac{-1}{2} - \frac{1}{2}i.$$

$$\rho = \sqrt{\left(\frac{-1}{2}\right)^2 + \left(\frac{1}{2}\right)^2}$$

$$\rho = \sqrt{\frac{1}{4} + \frac{1}{4}}$$

$$\rho = \sqrt{\frac{1}{2}}$$

$$\rho = \frac{1}{\sqrt{2}}.$$

$$\tan \theta = \frac{\frac{1}{2}}{\frac{-1}{2}}$$

$$\tan \theta = -1.$$

$$r_1, r_2 = \frac{1}{\sqrt{2}} \left(\cos \frac{3\pi}{4} \pm i \sin \frac{3\pi}{4} \right).$$

$$y_t^h = \left(\frac{1}{\sqrt{2}}\right)^t \left(C_1 \cos \frac{3\pi t}{4} + C_2 \sin \frac{3\pi t}{4} \right).$$

Solución particular:

$$y_t^p = k 2^{-t}.$$

$$y_{t+1} = k 2^{-(t+1)}$$

$$y_{t+1} = k 2^{-t-1}$$

$$y_{t+1} = \frac{k}{2} 2^{-t}.$$

$$y_{t+2} = k 2^{-(t+2)}$$

$$y_{t+2} = k 2^{-t-2}$$

$$y_{t+2} = \frac{k}{4} 2^{-t}.$$

$$2 \frac{k}{4} 2^{-t} + 2 \frac{k}{2} 2^{-t} + k 2^{-t} = 2^{-t}$$

$$\frac{k}{2} 2^{-t} + k 2^{-t} + k 2^{-t} = 2^{-t}$$

$$\left(\frac{k}{2} + k + k\right) 2^{-t} = 2^{-t}$$

$$\frac{5}{2} k 2^{-t} = 2^{-t}$$

$$\frac{5}{2} k = \frac{2^{-t}}{2^{-t}}$$

$$\frac{5}{2} k = 1$$

$$k = \frac{1}{\frac{5}{2}}$$

$$k = \frac{2}{5}.$$

$$y_t^p = \frac{2}{5} 2^{-t}.$$

Solución general:

$$y_t = y_t^h + y_t^p$$

$$y_t = \left(\frac{1}{\sqrt{2}}\right)^t \left(C_1 \cos \frac{3\pi t}{4} + C_2 \sin \frac{3\pi t}{4}\right) + \frac{2}{5} 2^{-t}.$$

Solución final:

$$y_0 = \left(\frac{1}{\sqrt{2}}\right)^0 \left(C_1 \cos \frac{3\pi \cdot 0}{4} + C_2 \sin \frac{3\pi \cdot 0}{4}\right) + \frac{2}{5} 2^{-0}$$

$$y_0 = 1 \left(C_1 \cos \frac{0}{4} + C_2 \sin \frac{0}{4}\right) + \frac{2}{5} \cdot 1$$

$$y_0 = (C_1 \cos 0 + C_2 \sin 0) + \frac{2}{5}$$

$$y_0 = (C_1 \cdot 1 + C_2 \cdot 0) + \frac{2}{5}$$

$$y_0 = (C_1 + 0) + \frac{2}{5}$$

$$0 = C_1 + \frac{2}{5}$$

$$C_1 = \frac{-2}{5}.$$

$$y_1 = \left(\frac{1}{\sqrt{2}}\right)^1 \left(\frac{-2}{5} \cos \frac{3\pi \cdot 1}{4} + C_2 \sin \frac{3\pi \cdot 1}{4}\right) + \frac{2}{5} 2^{-1}$$

$$y_1 = \frac{1}{\sqrt{2}} \left[\frac{-2}{5} \left(\frac{-\sqrt{2}}{2}\right) + C_2 \frac{\sqrt{2}}{2}\right] + \frac{1}{5}$$

$$y_1 = \frac{1}{\sqrt{2}} \left(\frac{\sqrt{2}}{5} + C_2 \frac{\sqrt{2}}{2}\right) + \frac{1}{5}$$

$$y_1 = \frac{1}{5} + \frac{1}{2} C_2 + \frac{1}{5}$$

$$y_1 = \frac{1}{2} C_2 + \frac{2}{5}$$

$$0 = \frac{1}{2} C_2 + \frac{2}{5}$$

$$\frac{1}{2} C_2 = -\frac{2}{5}$$

$$C_2 = -\frac{4}{5}$$

$$C_2 = -\frac{4}{5}.$$

$$y_t = \left(\frac{1}{\sqrt{2}}\right)^t \left[-\frac{2}{5} \cos \frac{3\pi t}{4} + \left(-\frac{4}{5}\right) \sin \frac{3\pi t}{4} \right] + \frac{2}{5} 2^{-t}.$$

$$\lim_{t \rightarrow +\infty} y_t = 0.$$

Por lo tanto, la solución converge a 0 y oscila (por el seno y coseno), ya que $\left|\frac{1}{\sqrt{2}}\right| < 1$ y $\left|\frac{1}{2}\right| < 1$.

Ejercicio 6.

Encontrar todas las soluciones de las siguientes ecuaciones:

(a) $y_{t+2} + 2y_{t+1} + y_t = 3^t$.

Solución homogénea:

$$y_t = r^t.$$

$$r^{t+2} + 2r^{t+1} + r^t = 0$$

$$r^t (r^2 + 2r + 1) = 0.$$

$$r_1, r_2 = \frac{-2 \pm \sqrt{2^2 - 4 \cdot 1 \cdot 1}}{2 \cdot 1}$$

$$r_1, r_2 = \frac{-2 \pm \sqrt{4 - 4}}{2}$$

$$r_1, r_2 = \frac{-2 \pm \sqrt{0}}{2}$$

$$r_1, r_2 = \frac{-2 \pm 0}{2}$$

$$r_1, r_2 = \frac{-2 + 0}{2} = \frac{-2}{2} = -1.$$

$$r_1, r_2 = \frac{-2 - 0}{2} = \frac{-2}{2} = -1.$$

$$y_t^h = (C_1 + C_2 t) (-1)^t.$$

Solución particular:

$$y_t^p = k 3^t.$$

$$y_{t+1} = k 3^{t+1}$$

$$y_{t+1} = 3k 3^t.$$

$$y_{t+2} = k 3^{t+2}$$

$$y_{t+2} = 9k 3^t.$$

$$9k 3^t + 2 \cdot 3k 3^t + k 3^t = 3^t$$

$$9k 3^t + 6k 3^t + k 3^t = 3^t$$

$$16k 3^t = 3^t$$

$$16k = \frac{3^t}{3^t}$$

$$16k = 1$$

$$k = \frac{1}{16}.$$

$$y_t^p = \frac{3^t}{16}.$$

Solución general:

$$y_t = y_t^h + y_t^p$$

$$y_t = (C_1 + C_2 t) (-1)^t + \frac{3^t}{16}.$$

$$(b) y_{t+2} - 5y_{t+1} - 6y_t = 2 * 3^t.$$

Solución homogénea:

$$y_t = r^t.$$

$$r^{t+2} - 5r^{t+1} - 6r^t = 0$$

$$r^t (r^2 - 5r - 6) = 0.$$

$$r_1, r_2 = \frac{-(-5) \pm \sqrt{(-5)^2 - 4*1*(-6)}}{2*1}$$

$$r_1, r_2 = \frac{5 \pm \sqrt{25+24}}{2}$$

$$r_1, r_2 = \frac{5 \pm \sqrt{49}}{2}$$

$$r_1, r_2 = \frac{5 \pm 7}{2}$$

$$r_1, r_2 = \frac{5+7}{2} = \frac{12}{2} = 6.$$

$$r_1, r_2 = \frac{5-7}{2} = \frac{-2}{2} = -1.$$

$$y_t^h = C_1 6^t + C_2 (-1)^t.$$

Solución particular:

$$y_t^p = k 3^t.$$

$$y_{t+1} = k 3^{t+1}$$

$$y_{t+1} = 3k 3^t.$$

$$y_{t+2} = k 3^{t+2}$$

$$y_{t+2} = 9k 3^t.$$

$$9k 3^t - 5 * 3k 3^t - 6k 3^t = 2 * 3^t$$

$$9k 3^t - 15k 3^t - 6k 3^t = 2 * 3^t$$

$$-12k 3^t = 2 * 3^t$$

$$-12k = 2 \frac{3^t}{3^t}$$

$$k = \frac{2}{-12}$$

$$k = \frac{-1}{6}.$$

$$y_t^p = \frac{-1}{6} 3^t.$$

Solución general:

$$y_t = y_t^h + y_t^p$$

$$y_t = C_1 6^t + C_2 (-1)^t - \frac{1}{6} 3^t.$$

$$(c) 3y_{t+2} + 9y_t = 3 \cdot 4^t.$$

Solución homogénea:

$$y_t = r^t.$$

$$3r^{t+2} + 9r^t = 0$$

$$3r^t (r^2 + 3) = 0.$$

$$r^2 + 3 = 0$$

$$r^2 = -3$$

$$\sqrt{r} = \sqrt{-3}$$

$$|r| = \sqrt{-1} \sqrt{-3}$$

$$r = \pm \sqrt{3}i.$$

$$\rho = \sqrt{(\sqrt{3})^2}$$

$$\rho = \sqrt{3}.$$

$$\tan \theta = \frac{\sqrt{3}}{0}.$$

$$r_1, r_2 = \sqrt{3} \left(\cos \frac{\pi}{2} \pm i \sin \frac{\pi}{2} \right).$$

$$y_t^h = (\sqrt{3})^t \left(C_1 \cos \frac{\pi t}{2} + C_2 \sin \frac{\pi t}{2} \right).$$

Solución particular:

$$y_t^p = k 4^t.$$

$$y_{t+1} = k 4^{t+1}$$

$$y_{t+1} = 4k 4^t.$$

$$y_{t+2} = k 4^{t+2}$$

$$y_{t+2} = 16k 4^t.$$

$$3 \cdot 16k 4^t + 9k 4^t = 3 \cdot 4^t$$

$$16k 4^t + 3k 4^t = 4^t$$

$$19k 4^t = 4^t$$

$$19k = \frac{4^t}{4^t}$$

$$19k = 1$$

$$k = \frac{1}{19},$$

$$y_t^p = \frac{4^t}{19}.$$

Solución general:

$$y_t = y_t^h + y_t^p$$

$$y_t = (\sqrt{3})^t (C_1 \cos \frac{\pi t}{2} + C_2 \sin \frac{\pi t}{2}) + \frac{4^t}{19}.$$

$$(d) y_{t+2} + 5y_{t+1} + 2y_t = e^t.$$

Solución homogénea:

$$y_t = r^t.$$

$$r^{t+2} + 5r^{t+1} + 2r^t = 0$$

$$r^t (r^2 + 5r + 2) = 0.$$

$$r_1, r_2 = \frac{-5 \pm \sqrt{5^2 - 4 \cdot 1 \cdot 2}}{2 \cdot 1}$$

$$r_1, r_2 = \frac{-5 \pm \sqrt{25-8}}{2}$$

$$r_1, r_2 = \frac{-5 \pm \sqrt{17}}{2}$$

$$r_1 = \frac{-5 + \sqrt{17}}{2}.$$

$$r_2 = \frac{-5 - \sqrt{17}}{2}.$$

$$y_t^h = C_1 \left(\frac{-5 + \sqrt{17}}{2} \right)^t + C_2 \left(\frac{-5 - \sqrt{17}}{2} \right)^t.$$

Solución particular:

$$y_t^p = k e^t.$$

$$y_{t+1} = k e^{t+1}$$

$$y_{t+1} = e k e^t.$$

$$y_{t+2} = k e^{t+2}$$

$$y_{t+2} = e^2 k e^t.$$

$$e^2 k e^t + 5 e k e^t + 2 k e^t = e^t$$

$$(e^2 + 5e + 2) k e^t = e^t$$

$$(e^2 + 5e + 2) k = \frac{e^t}{e^t}$$

$$(e^2 + 5e + 2) k = 1$$

$$k = \frac{1}{e^2 + 5e + 2}.$$

$$y_t^p = \frac{e^t}{e^2 + 5e + 2}.$$

Solución general:

$$y_t = y_t^h + y_t^p$$

$$y_t = C_1 \left(\frac{-5 + \sqrt{17}}{2} \right)^t + C_2 \left(\frac{-5 - \sqrt{17}}{2} \right)^t + \frac{e^t}{e^2 + 5e + 2}.$$

(e) $y_{t+2} - 2y_{t+1} + 5y_t = t.$

Solución homogénea:

$$y_t = r^t.$$

$$r^{t+2} - 2r^{t+1} + 5r^t = 0$$

$$r^t (r^2 - 2r + 5) = 0.$$

$$r_1, r_2 = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \cdot 1 \cdot 5}}{2 \cdot 1}$$

$$r_1, r_2 = \frac{2 \pm \sqrt{4 - 20}}{2}$$

$$r_1, r_2 = \frac{2 \pm \sqrt{-16}}{2}$$

$$r_1, r_2 = \frac{2 \pm 4i}{2}$$

$$r_1 = \frac{2 + 4i}{2} = 1 + 2i.$$

$$r_2 = \frac{2 - 4i}{2} = 1 - 2i.$$

$$\rho = \sqrt{1^2 + 2^2}$$

$$\rho = \sqrt{1 + 4}$$

$$\rho = \sqrt{5}.$$

$$\tan \theta = \frac{2}{1}$$

$$\tan \theta = 2.$$

$$r_1, r_2 = \sqrt{5} [\cos (\arctan 2) \pm i \sin (\arctan 2)].$$

$$y_t^h = (\sqrt{5})^t [C_1 \cos (t \arctan 2) + C_2 \sin (t \arctan 2)].$$

Solución particular:

$$y_t^p = k_1 t + k_2.$$

$$y_{t+1} = k_1 (t + 1) + k_2$$

$$y_{t+1} = k_1 t + k_1 + k_2.$$

$$y_{t+2} = k_1 (t + 2) + k_2$$

$$y_{t+2} = k_1 t + 2k_1 + k_2.$$

$$k_1 t + 2k_1 + k_2 - 2(k_1 t + k_1 + k_2) + 5(k_1 t + k_2) = t$$

$$4k_1 t + 4k_2 = t.$$

$$4k_1 = t$$

$$k_1 = \frac{1}{4} t.$$

$$4k_2 = 0$$

$$k_2 = \frac{0}{4}$$

$$k_2 = 0.$$

$$y_t^p = \frac{1}{4} t.$$

Solución general:

$$y_t = y_t^h + y_t^p$$

$$y_t = (\sqrt{5})^t [C_1 \cos(t \arctan 2) + C_2 \sin(t \arctan 2)] + \frac{1}{4} t.$$

(f) $y_{t+2} - 2y_{t+1} + 5y_t = 4 + 2t.$

Solución homogénea:

$$y_t = r^t.$$

$$r^{t+2} - 2r^{t+1} + 5r^t = 0$$

$$r^t (r^2 - 2r + 5) = 0.$$

$$r_1, r_2 = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \cdot 1 \cdot 5}}{2 \cdot 1}$$

$$r_1, r_2 = \frac{2 \pm \sqrt{4-20}}{2}$$

$$r_1, r_2 = \frac{2 \pm \sqrt{-16}}{2}$$

$$r_1, r_2 = \frac{2 \pm 4i}{2}$$

$$r_1 = \frac{2+4i}{2} = 1 + 2i.$$

$$r_2 = \frac{2-4i}{2} = 1 - 2i.$$

$$\rho = \sqrt{1^2 + 2^2}$$

$$\rho = \sqrt{1 + 4}$$

$$\rho = \sqrt{5}.$$

$$\tan \theta = \frac{2}{1}$$

$$\tan \theta = 2.$$

$$r_1, r_2 = \sqrt{5} [\cos (\arctan 2) \pm i \operatorname{sen} (\arctan 2)].$$

$$y_t^h = (\sqrt{5})^t [C_1 \cos (t \arctan 2) + C_2 \operatorname{sen} (t \arctan 2)].$$

Solución particular:

$$y_t^p = k_1 t + k_2.$$

$$y_{t+1} = k_1 (t + 1) + k_2$$

$$y_{t+1} = k_1 t + k_1 + k_2.$$

$$y_{t+2} = k_1 (t + 2) + k_2$$

$$y_{t+2} = k_1 t + 2k_1 + k_2.$$

$$k_1 t + 2k_1 + k_2 - 2(k_1 t + k_1 + k_2) + 5(k_1 t + k_2) = 4 + 2t$$

$$4k_1 t + 4k_2 = 4 + 2t.$$

$$4k_1 = 2t$$

$$k_1 = \frac{2}{4} t$$

$$k_1 = \frac{1}{2} t$$

$$4k_2 = 4$$

$$k_2 = \frac{4}{4}$$

$$k_2 = 1.$$

$$y_t^p = \frac{1}{2} t + 1.$$

Solución general:

$$y_t = y_t^h + y_t^p$$

$$y_t = (\sqrt{5})^t [C_1 \cos (t \arctan 2) + C_2 \operatorname{sen} (t \arctan 2)] + \frac{1}{2} t + 1.$$

$$(g) \ y_{t+2} + 5y_{t+1} + 2y_t = 18 + 6t + 8t^2.$$

Solución homogénea:

$$y_t = r^t.$$

$$r^{t+2} + 5r^{t+1} + 2r^t = 0$$

$$r^t (r^2 + 5r + 2) = 0.$$

$$r_1, r_2 = \frac{-5 \pm \sqrt{5^2 - 4 \cdot 1 \cdot 2}}{2 \cdot 1}$$

$$r_1, r_2 = \frac{-5 \pm \sqrt{25 - 8}}{2}$$

$$r_1, r_2 = \frac{-5 \pm \sqrt{17}}{2}$$

$$r_1 = \frac{-5 + \sqrt{17}}{2}.$$

$$r_2 = \frac{-5 - \sqrt{17}}{2}.$$

$$y_t^h = C_1 \left(\frac{-5 + \sqrt{17}}{2} \right)^t + C_2 \left(\frac{-5 - \sqrt{17}}{2} \right)^t.$$

Solución particular:

$$y_t^p = k_1 t^2 + k_2 t + k_3.$$

$$y_{t+1} = k_1 (t+1)^2 + k_2 (t+1) + k_3$$

$$y_{t+1} = k_1 (t^2 + 2t + 1) + k_2 t + k_2 + k_3$$

$$y_{t+1} = k_1 t^2 + 2t k_1 + k_1 + k_2 t + k_2 + k_3$$

$$y_{t+1} = k_1 t^2 + (2k_1 + k_2) t + (k_1 + k_2 + k_3).$$

$$y_{t+2} = k_1 (t+2)^2 + k_2 (t+2) + k_3$$

$$y_{t+2} = k_1 (t^2 + 4t + 4) + k_2 t + 2k_2 + k_3$$

$$y_{t+2} = k_1 t^2 + 4t k_1 + 4k_1 + k_2 t + 2k_2 + k_3$$

$$y_{t+2} = k_1 t^2 + (4k_1 + k_2) t + (4k_1 + 2k_2 + k_3).$$

$$k_1 t^2 + (4k_1 + k_2) t + (4k_1 + 2k_2 + k_3) + 5 [k_1 t^2 + (2k_1 + k_2) t + (k_1 + k_2 + k_3)] + 2$$

$$(k_1 t^2 + k_2 t + k_3) = 18 + 6t + 8t^2$$

$$k_1 t^2 + (4k_1 + k_2) t + (4k_1 + 2k_2 + k_3) + 5k_1 t^2 + 5(2k_1 + k_2) t + 5(k_1 + k_2 + k_3) +$$

$$2k_1 t^2 + 2k_2 t + 2k_3 = 18 + 6t + 8t^2$$

$$k_1 t^2 + (4k_1 + k_2) t + (4k_1 + 2k_2 + k_3) + 5k_1 t^2 + (10k_1 + 5k_2) t + (5k_1 + 5k_2 + 5k_3) +$$

$$2k_1 t^2 + 2k_2 t + 2k_3 = 18 + 6t + 8t^2$$

$$8k_1 t^2 + (14k_1 + 8k_2) t + (9k_1 + 7k_2 + 8k_3) = 18 + 6t + 8t^2.$$

$$8k_1 = 8$$

$$k_1 = \frac{8}{8}$$

$$k_1 = 1.$$

$$14k_1 + 8k_2 = 6$$

$$14 * 1 + 8k_2 = 6$$

$$14 + 8k_2 = 6$$

$$8k_2 = 6 - 14$$

$$8k_2 = -8$$

$$k_2 = \frac{-8}{8}$$

$$k_2 = -1.$$

$$9k_1 + 7k_2 + 8k_3 = 18$$

$$9 * 1 + 7(-1) + 8k_3 = 18$$

$$9 - 7 + 8k_3 = 18$$

$$2 + 8k_3 = 18$$

$$8k_3 = 18 - 2$$

$$8k_3 = 16$$

$$k_3 = \frac{16}{8}$$

$$k_3 = 2.$$

$$y_t^p = t^2 - t + 2.$$

Solución general:

$$y_t = y_t^h + y_t^p$$

$$y_t = C_1 \left(\frac{-5 + \sqrt{17}}{2} \right)^t + C_2 \left(\frac{-5 - \sqrt{17}}{2} \right)^t + t^2 - t + 2.$$

$$(h) y_{t+2} + 5y_{t+1} + 2y_t = e^t + 18 + 6t + 8t^2.$$

Solución homogénea:

$$y_t = r^t.$$

$$r^{t+2} + 5r^{t+1} + 2r^t = 0$$

$$r^t (r^2 + 5r + 2) = 0.$$

$$r_1, r_2 = \frac{-5 \pm \sqrt{5^2 - 4 \cdot 1 \cdot 2}}{2 \cdot 1}$$

$$r_1, r_2 = \frac{-5 \pm \sqrt{25 - 8}}{2}$$

$$r_1, r_2 = \frac{-5 \pm \sqrt{17}}{2}$$

$$r_1 = \frac{-5 + \sqrt{17}}{2}.$$

$$r_2 = \frac{-5 - \sqrt{17}}{2}.$$

$$y_t^h = C_1 \left(\frac{-5 + \sqrt{17}}{2} \right)^t + C_2 \left(\frac{-5 - \sqrt{17}}{2} \right)^t.$$

Solución particular:

$$y_t^p = y_t^{p_1} + y_t^{p_2}$$

$$y_t^p = e^t + k_1 t^2 + k_2 t + k_3.$$

$$y_t^{p_1} = k e^t.$$

$$y_{t+1} = k e^{t+1}$$

$$y_{t+1} = e k e^t.$$

$$y_{t+2} = k e^{t+2}$$

$$y_{t+2} = e^2 k e^t.$$

$$e^2 k e^t + 5 e k e^t + 2 k e^t = e^t$$

$$(e^2 + 5e + 2) k e^t = e^t$$

$$(e^2 + 5e + 2) k = \frac{e^t}{e^t}$$

$$(e^2 + 5e + 2) k = 1$$

$$k = \frac{1}{e^2 + 5e + 2}.$$

$$y_t^{p_1} = \frac{e^t}{e^2 + 5e + 2}.$$

$$y_t^{p_2} = k_1 t^2 + k_2 t + k_3.$$

$$y_{t+1} = k_1 (t + 1)^2 + k_2 (t + 1) + k_3$$

$$y_{t+1} = k_1 (t^2 + 2t + 1) + k_2 t + k_2 + k_3$$

$$y_{t+1} = k_1 t^2 + 2t k_1 + k_1 + k_2 t + k_2 + k_3$$

$$y_{t+1} = k_1 t^2 + (2k_1 + k_2) t + (k_1 + k_2 + k_3).$$

$$y_{t+2} = k_1 (t + 2)^2 + k_2 (t + 2) + k_3$$

$$y_{t+2} = k_1 (t^2 + 4t + 4) + k_2 t + 2k_2 + k_3$$

$$y_{t+2} = k_1 t^2 + 4t k_1 + 4k_1 + k_2 t + 2k_2 + k_3$$

$$y_{t+2} = k_1 t^2 + (4k_1 + k_2) t + (4k_1 + 2k_2 + k_3).$$

$$k_1 t^2 + (4k_1 + k_2) t + (4k_1 + 2k_2 + k_3) + 5 [k_1 t^2 + (2k_1 + k_2) t + (k_1 + k_2 + k_3)] + 2 (k_1 t^2 + k_2 t + k_3) = 18 + 6t + 8t^2$$

$$k_1 t^2 + (4k_1 + k_2) t + (4k_1 + 2k_2 + k_3) + 5k_1 t^2 + 5(2k_1 + k_2) t + 5(k_1 + k_2 + k_3) + 2k_1 t^2 + 2k_2 t + 2k_3 = 18 + 6t + 8t^2$$

$$k_1 t^2 + (4k_1 + k_2) t + (4k_1 + 2k_2 + k_3) + 5k_1 t^2 + (10k_1 + 5k_2) t + (5k_1 + 5k_2 + 5k_3) + 2k_1 t^2 + 2k_2 t + 2k_3 = 18 + 6t + 8t^2$$

$$8k_1 t^2 + (14k_1 + 8k_2) t + (9k_1 + 7k_2 + 8k_3) = 18 + 6t + 8t^2.$$

$$8k_1 = 8$$

$$k_1 = \frac{8}{8}$$

$$k_1 = 1.$$

$$14k_1 + 8k_2 = 6$$

$$14 * 1 + 8k_2 = 6$$

$$14 + 8k_2 = 6$$

$$8k_2 = 6 - 14$$

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$$9 - 7 + 8k_3 = 18$$

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$$8k_3 = 18 - 2$$

$$8k_3 = 16$$

$$k_3 = \frac{16}{8}$$

$$k_3 = 2.$$

$$y_t^{p_2} = t^2 - t + 2.$$

$$y_t^p = \frac{e^t}{e^2 + 5e + 2} + t^2 - t + 2.$$

Solución general:

$$y_t = y_t^h + y_t^p$$

$$y_t = C_1 \left(\frac{-5 + \sqrt{17}}{2} \right)^t + C_2 \left(\frac{-5 - \sqrt{17}}{2} \right)^t + \frac{e^t}{e^2 + 5e + 2} + t^2 - t + 2.$$

Ejercicio 7 (*).

Resolver la siguiente ecuación en diferencia de orden 2:

$$y_{t+2} - 3y_{t+1} - 4y_t = 2^t + t^3.$$

Se parte de saber que la solución general de una ecuación en diferencias son las soluciones de la ecuación homogénea asociada a ésta más la solución particular:

$$y_t = y_t^h + y_t^p.$$

Siendo así, en primer lugar, se buscan las soluciones de la ecuación homogénea asociada a la ecuación en diferencias:

$$y_{t+2} - 3y_{t+1} - 4y_t = 0.$$

Sea $y_t = r^t$, se tiene:

$$\begin{aligned} r^{t+2} - 3r^{t+1} - 4r^t &= 0 \\ r^t (r^2 - 3r - 4) &= 0. \end{aligned}$$

Resolviendo mediante el método de Bhaskara, se hallan las soluciones de la ecuación homogénea:

$$\begin{aligned} r_1, r_2 &= \frac{-(-3) \pm \sqrt{(-3)^2 - 4 \cdot 1 \cdot (-4)}}{2 \cdot 1} \\ r_1, r_2 &= \frac{3 \pm \sqrt{9+16}}{2} \\ r_1, r_2 &= \frac{3 \pm \sqrt{25}}{2} \\ r_1, r_2 &= \frac{3 \pm 5}{2} \\ r_1 &= \frac{3+5}{2} = \frac{8}{2} = 4. \\ r_2 &= \frac{3-5}{2} = \frac{-2}{2} = -1. \end{aligned}$$

Entonces, la solución homogénea es la siguiente:

$$y_t^h = C_1 4^t + C_2 (-1)^t.$$

En segundo lugar, dado que 1 no es raíz de la ecuación $r^2 - 3r - 4 = 0$, se propone la siguiente solución particular:

$$y_t^p = k_1 2^t + k_2 t^3 + k_3 t^2 + k_4 t + k_5.$$

Resolviendo en la ecuación en diferencias, se tiene:

$$\begin{aligned} &[k_1 2^{t+2} + k_2 (t+2)^3 + k_3 (t+2)^2 + k_4 (t+2) + k_5] - \\ &3 [k_1 2^{t+1} + k_2 (t+1)^3 + k_3 (t+1)^2 + k_4 (t+1) + k_5] - \\ &4 [k_1 2^t + k_2 t^3 + k_3 t^2 + k_4 t + k_5] = 2^t + t^3 \end{aligned}$$

$$[k_1 2^{t+2} + k_2 (t^3 + 6t^2 + 12t + 8) + k_3 (t^2 + 4t + 4) + k_4 t + 2k_4 + k_5] -$$

$$3 [k_1 2^{t+1} + k_2 (t^3 + 3t^2 + 3t + 1) + k_3 (t^2 + 2t + 1) + k_4 t + k_4 + k_5] -$$

$$(4k_1 2^t + 4k_2 t^3 + 4k_3 t^2 + 4k_4 t + 4k_5) = 2^t + t^3$$

$$k_1 2^{t+2} + k_2 t^3 + 6k_2 t^2 + 12k_2 t + 8k_2 + k_3 t^2 + 4k_3 t + 4k_3 + k_4 t + 2k_4 + k_5 -$$

$$3 [k_1 2^{t+1} + k_2 t^3 + 3k_2 t^2 + 3k_2 t + k_2 + k_3 t^2 + 2k_3 t + k_3 + k_4 t + k_4 + k_5] -$$

$$4k_1 2^t - 4k_2 t^3 - 4k_3 t^2 - 4k_4 t - 4k_5 = 2^t + t^3$$

$$k_1 2^{t+2} + k_2 t^3 + 6k_2 t^2 + 12k_2 t + 8k_2 + k_3 t^2 + 4k_3 t + 4k_3 + k_4 t + 2k_4 + k_5 -$$

$$(3k_1 2^{t+1} + 3k_2 t^3 + 9k_2 t^2 + 9k_2 t + 3k_2 + 3k_3 t^2 + 6k_3 t + 3k_3 + 3k_4 t + 3k_4 + 3k_5) -$$

$$4k_1 2^t - 4k_2 t^3 - 4k_3 t^2 - 4k_4 t - 4k_5 = 2^t + t^3$$

$$k_1 2^{t+2} + k_2 t^3 + 6k_2 t^2 + 12k_2 t + 8k_2 + k_3 t^2 + 4k_3 t + 4k_3 + k_4 t + 2k_4 + k_5 -$$

$$3k_1 2^{t+1} - 3k_2 t^3 - 9k_2 t^2 - 9k_2 t - 3k_2 - 3k_3 t^2 - 6k_3 t - 3k_3 - 3k_4 t - 3k_4 - 3k_5 -$$

$$4k_1 2^t - 4k_2 t^3 - 4k_3 t^2 - 4k_4 t - 4k_5 = 2^t + t^3$$

$$-6k_1 2^t - 6k_2 t^3 - (3k_2 + 6k_3) t^2 + (3k_2 - 2k_3 - 6k_4) t + (5k_2 + k_3 - k_4 - 6k_5) = 2^t + t^3$$

Luego, se tiene que las constantes k_1 , k_2 , k_3 , k_4 y k_5 , respectivamente, son iguales a:

$$-6k_1 2^t = 2^t$$

$$k_1 = \frac{2^t}{-6 \cdot 2^t}$$

$$k_1 = \frac{-1}{6}.$$

$$-6k_2 t^3 = t^3$$

$$k_2 = \frac{t^3}{-6t^3}$$

$$k_2 = \frac{-1}{6}.$$

$$-(3k_2 + 6k_3) t^2 = 0$$

$$3k_2 + 6k_3 = \frac{0}{-t^2}$$

$$3k_2 + 6k_3 = 0$$

$$6k_3 = -3k_2$$

$$k_3 = \frac{-3k_2}{6}$$

$$k_3 = \frac{-1}{2} k_2$$

$$k_3 = \frac{-1}{2} \left(\frac{-1}{6} \right)$$

$$k_3 = \frac{1}{12}.$$

$$(3k_2 - 2k_3 - 6k_4) t = 0$$

$$3k_2 - 2k_3 - 6k_4 = \frac{0}{t}$$

$$3k_2 - 2k_3 - 6k_4 = 0$$

$$6k_4 = 3k_2 - 2k_3$$

$$k_4 = \frac{3k_2 - 2k_3}{6}$$

$$k_4 = \frac{1}{2} k_2 - \frac{1}{3} k_3$$

$$k_4 = \frac{1}{2} \left(\frac{-1}{6} \right) - \frac{1}{3} \frac{1}{12}$$

$$k_4 = \frac{-1}{12} - \frac{1}{36}$$

$$k_4 = \frac{-1}{9}.$$

$$5k_2 + k_3 - k_4 - 6k_5 = 0$$

$$6k_5 = 5k_2 + k_3 - k_4$$

$$k_5 = \frac{5k_2 + k_3 - k_4}{6}$$

$$k_5 = \frac{5}{6}k_2 + \frac{1}{6}k_3 - \frac{1}{6}k_4$$

$$k_5 = \frac{5}{6}\left(\frac{-1}{6}\right) + \frac{1}{6}\frac{1}{12} - \frac{1}{6}\left(\frac{-1}{9}\right)$$

$$k_5 = \frac{-5}{36} + \frac{1}{72} + \frac{1}{54}$$

$$k_5 = \frac{-23}{216}.$$

Entonces, la solución particular es la siguiente:

$$y_t^p = \frac{-1}{6} 2^t - \frac{1}{6} t^3 + \frac{1}{12} t^2 - \frac{1}{9} t - \frac{23}{216}.$$

Por lo tanto, la solución general de la ecuación en diferencias de orden 2 es:

$$y_t = C_1 4^t + C_2 (-1)^t - \frac{1}{6} 2^t - \frac{1}{6} t^3 + \frac{1}{12} t^2 - \frac{1}{9} t - \frac{23}{216}.$$